

Bootstrap

1. Introduction to Bootstrap

Bootstrap is a **popular front-end CSS framework** used to build **responsive, mobile-first web applications** quickly and efficiently.

Key Points

- Developed by Twitter
 - Open-source framework
 - Based on **HTML, CSS, and JavaScript**
 - Uses **predefined CSS classes**
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2. Why Use Bootstrap?

- Saves development time
 - Provides responsive design by default
 - Consistent UI across browsers
 - No need to write CSS from scratch
 - Easy to learn and use
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3. Bootstrap Versions

- Bootstrap 3 – Older, uses float-based grid
- Bootstrap 4 – Flexbox-based grid
- **Bootstrap 5 – Latest, no jQuery dependency**

This documentation uses **Bootstrap 5**.

4. Including Bootstrap in HTML (CDN Method)

4.1 Bootstrap CSS CDN

```
<link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.mi  
n.css" rel="stylesheet">
```

4.2 Bootstrap JS CDN (Bundle)

```
<script  
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
```

The bundle includes **Popper.js**, required for dropdowns, tooltips, and modals.

5. Bootstrap Container

Containers are used to **wrap content and apply responsive padding**.

Types of Containers

Class	Description
container	Fixed-width container
container-fluid	Full-width container

Example

```
<div class="container">  
  <h1>Hello Bootstrap</h1>  
</div>
```

6. Bootstrap Grid System (Most Important)

Bootstrap uses a **12-column responsive grid system**.

Grid Rules

- Must be inside a container
- Rows contain columns
- Columns must be inside rows

Basic Structure

```
<div class="container">  
  <div class="row">
```

```
<div class="col-4">Column 1</div>
<div class="col-4">Column 2</div>
<div class="col-4">Column 3</div>
</div>
</div>
```

Responsive Breakpoints

Class Prefix	Screen Size
col-sm	≥576px
col-md	≥768px
col-lg	≥992px
col-xl	≥1200px

Example

```
<div class="col-sm-12 col-md-6 col-lg-4">Content</div>
```

7. Typography

Bootstrap provides predefined typography styles.

```
<h1 class="display-4">Heading</h1>
<p class="lead">Lead paragraph</p>
<p class="text-muted">Muted text</p>
```

8. Colors and Backgrounds

Text Colors

```
<p class="text-primary">Primary</p>
<p class="text-success">Success</p>
<p class="text-danger">Danger</p>
```

Background Colors

```
<div class="bg-warning p-3">Warning Background</div>
```

9. Spacing Utilities

Bootstrap uses margin (**m**) and padding (**p**) utilities.

Syntax

```
{property}{side}-{size}
```

Example:

```
<div class="mt-4 mb-3 p-2">Spacing Example</div>
```

10. Buttons

```
<button class="btn btn-primary">Primary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-outline-danger">Danger</button>
```

11. Alerts

```
<div class="alert alert-success">Operation Successful</div>
<div class="alert alert-danger">Error Occurred</div>
```

12. Cards

```
<div class="card" style="width: 18rem;">
  <div class="card-body">
```

```
<h5 class="card-title">Card Title</h5>
<p class="card-text">Card content</p>
<a href="#" class="btn btn-primary">Go</a>
</div>
</div>
```

13. Navbar

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#">Brand</a>
  </div>
</nav>
```

Navbar toggle requires **Bootstrap JS CDN**.

14. Forms

```
<div class="mb-3">
  <label class="form-label">Email</label>
  <input type="email" class="form-control">
</div>
```

Common Form Classes

- form-control
- form-label
- form-check

15. Tables

```
<table class="table table-bordered table-striped">
  <thead class="table-dark">
    <tr><th>ID</th><th>Name</th></tr>
  </thead>
</table>
```

16. Bootstrap JS Components

Components requiring JS:

- Modal
- Dropdown
- Collapse
- Navbar toggle

Modal Example

```
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#m1">Open</button>
```
